

Last login: Wed Aug 12 12:14:15 on ttys001
harsha@Harshas-MacBook-Air:~ % /var/folders/g6/2h7p_f2x2qbcf0_zwd7zs4y00000gn/T/orbstack-open-terminal_9F57F088.sh ; exit;

OrbStack update available!
Run "orb update" to update.

Updates include improvements, features, and fixes.

```
harsha@Harsha:~$ nano Makefile
harsha@Harsha:~$ make hello
make: *** No rule to make target 'src/hello.c', needed by 'hello'. Stop.
harsha@Harsha:~$ cd ~/os-lab
harsha@Harsha:~/os-lab$ pwd
ls
/home/harsha/os-lab
bin hello include Makefile README.md src
harsha@Harsha:~/os-lab$ ls src
harsha@Harsha:~/os-lab$ nano src/hello.c
harsha@Harsha:~/os-lab$ make hello
make: *** No rule to make target 'hello'. Stop.
harsha@Harsha:~/os-lab$ ./bin/hello
-bash: ./bin/hello: No such file or directory
harsha@Harsha:~/os-lab$ ls -l src/
total 4
-rw-r--r-- 1 harsha harsha 88 Aug 12 12:17 hello.c
harsha@Harsha:~/os-lab$ gcc src/hello.c -o bin/hello
harsha@Harsha:~/os-lab$ ls -l bin/
total 16
-rwxr-xr-x 1 harsha harsha 70312 Aug 12 12:18 hello
harsha@Harsha:~/os-lab$ ./bin/hello
Hello from Linux!
harsha@Harsha:~/os-lab$ make hello
make: *** No rule to make target 'hello'. Stop.
harsha@Harsha:~/os-lab$ ls -l src/
total 4
-rw-r--r-- 1 harsha harsha 88 Aug 12 12:17 hello.c
harsha@Harsha:~/os-lab$ cat Makefile
harsha@Harsha:~/os-lab$ make hello
make: *** No rule to make target 'hello'. Stop.
harsha@Harsha:~/os-lab$ nano Makefile
harsha@Harsha:~/os-lab$ ls -l src/
total 4
-rw-r--r-- 1 harsha harsha 88 Aug 12 12:17 hello.c
harsha@Harsha:~/os-lab$ make hello
gcc -Wall -Wextra -g src/hello.c -o bin/hello
harsha@Harsha:~/os-lab$ ./bin/hello
Hello from Linux!
harsha@Harsha:~/os-lab$
```

OrbStack update available!
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```
harsha@Harsha:~$ nano Makefile
harsha@Harsha:~$ make hello
make: *** No rule to make target 'src/hello.c', needed by 'hello'. Stop.
harsha@Harsha:~$ cd ~/os-lab
harsha@Harsha:~/os-lab$ pwd
ls
/home/harsha/os-lab
bin build include Makefile README.md src
harsha@Harsha:~/os-lab$ ls src
harsha@Harsha:~/os-lab$ nano src/hello.c
harsha@Harsha:~/os-lab$ make hello
make: *** No rule to make target 'hello'. Stop.
harsha@Harsha:~/os-lab$ ./bin/hello
-bash: ./bin/hello: No such file or directory
harsha@Harsha:~/os-lab$ ls -l src/
total 4
-rw-r--r-- 1 harsha harsha 88 Aug 12 12:17 hello.c
harsha@Harsha:~/os-lab$ gcc src/hello.c -o bin/hello
harsha@Harsha:~/os-lab$ ls -l bin/
total 16
-rwxr-xr-x 1 harsha harsha 78312 Aug 12 12:18 hello
harsha@Harsha:~/os-lab$ ./bin/hello
Hello from Linux!
harsha@Harsha:~/os-lab$ make hello
make: *** No rule to make target 'hello'. Stop.
harsha@Harsha:~/os-lab$ ls -l src/
total 4
-rw-r--r-- 1 harsha harsha 88 Aug 12 12:17 hello.c
harsha@Harsha:~/os-lab$ cat Makefile
harsha@Harsha:~/os-lab$ make hello
gcc -Wall -Wextra -g src/hello.c -o bin/hello
harsha@Harsha:~/os-lab$ ./bin/hello
Hello from Linux!
harsha@Harsha:~/os-lab$ mkdir -p src
nano src/fork_demo.c
harsha@Harsha:~/os-lab$ gcc src/fork_demo.c -o bin/fork_demo
harsha@Harsha:~/os-lab$ ./bin/fork_demo
Parent process: PID = 649, Child PID = 650
Child process: PID = 650, Parent PID = 649
harsha@Harsha:~/os-lab$ pidof pid = fork();
-bash: syntax error near unexpected token '('
harsha@Harsha:~/os-lab$ nano src/exec_demo.c
harsha@Harsha:~/os-lab$ gcc src/exec_demo.c -o bin/exec_demo
harsha@Harsha:~/os-lab$ ./bin/exec_demo
Before exec()
total 4
drwxr-xr-x 1 harsha harsha 46 Aug 12 12:26 bin
drwxr-xr-x 1 harsha harsha 0 Aug 12 12:15 build
drwxr-xr-x 1 harsha harsha 0 Aug 12 12:15 include
-rw-r--r-- 1 harsha harsha 110 Aug 12 12:19 Makefile
-rw-r--r-- 1 harsha harsha 8 Aug 12 12:15 README.md
drwxr-xr-x 1 harsha harsha 58 Aug 12 12:26 src
harsha@Harsha:~/os-lab$ strace ./bin/fork_demo
execve("./bin/fork_demo", ["/bin/fork_demo"], 0xffffd1b2e10 /* 26 vars */) = 0
brk(NULL) = 0xash009d7000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffffa674c000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
```

harsha — harsha@Harsha: ~/os-lab — orb - sh -e /var/folders/g6/2h7p_f2x2qbcf0_xwd7zs4y00000gn/T/orbstack-open-terminal_9F57F0B8.sh — 241x75

```
mmap(NULL, 13239, PROT_READ, MAP_PRIVATE, 3, 0) = 0xfffffa6748000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\340\206\2\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0755, st_size=1722928, ...}) = 0
mmap(NULL, 1892240, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xfffffa6548000
mmap(0xfffffa658000, 32768, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xfffffa658000
munmap(0xfffffa670000, 32656) = 0
mprotect(0xfffffa6ea000, 77824, PROT_NONE) = 0
mmap(0xfffffa6fd000, 28480, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x19d000) = 0xfffffa66fd000
mmap(0xfffffa6702000, 49040, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xfffffa6702000
close(3) = 0
set_tid_address(0xfffffa674cf90) = 661
set_robust_list(0xfffffa674cfa0, 24) = 0
rseq(0xfffffa674d5e0, 0x20, 0, 0xd428bc00) = 0
mprotect(0xfffffa66fd000, 12288, PROT_READ) = 0
mprotect(0xaaead233f000, 4096, PROT_READ) = 0
mprotect(0xfffffa6704000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0xfffffa6748000, 13239) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0xfffffa674cf90) = 662
getpid() = 661
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x1), ...}) = 0
getrandom(child_pid: PID = 662, Parent PID = 661) = 0
"\x9f\x2b\x18\xeb\x07\x46\x1a\x0e" = 8
brk(NULL) = 0xaab0b9d7000
brk(0xaab0b9f8000) = 0xaab0b9f8000
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=662, si_uid=501, si_status=0, si_etime=0, si_stime=0} ---
write(1, "Parent process: PID = 661, Child...., 43Parent process: PID = 661, Child PID = 662
") = 43
exit_group(0) = ?
+++ exited with 0 +++
(harsha@Harsha:~/os-lab$ strace ./bin/fork_demo
execve("./bin/fork_demo", ["/bin/fork_demo"], 0xffffd10d02c0 /* 26 vars */) = 0
brk(NULL) = 0xaab0bdc4000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff91e79000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=13239, ...}) = 0
mmap(NULL, 13239, PROT_READ, MAP_PRIVATE, 3, 0) = 0xffff91e75000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\340\206\2\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0755, st_size=1722928, ...}) = 0
mmap(NULL, 1892240, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xffff91c80000
mmap(0xffff91c80000, 1826704, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xffff91c80000
munmap(0xffff91c75000, 45056) = 0
munmap(0xffff91c8000, 28368) = 0
mprotect(0xffff91e1a000, 77824, PROT_NONE) = 0
mmap(0xffff91e20000, 28480, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x19d000) = 0xffff91e2d000
mmap(0xffff91e32000, 49040, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xffff91e32000
close(3) = 0
set_tid_address(0xffff91e79f90) = 666
set_robust_list(0xffff91e79fa0, 24) = 0
rseq(0xffff91e7a5e0, 0x20, 0, 0xd428bc00) = 0
mprotect(0xffff91e2d000, 12288, PROT_READ) = 0
mprotect(0xaaead40f000, 4096, PROT_READ) = 0
mprotect(0xffff91e81000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0xffff91e75000, 13239) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0xffff91e79f90) = 667
getpid() = 666
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x1), ...}) = 0
getrandom("\xf1\x9b\x08\xe0\x25\x41\xfc\x47", 8, GRND_NONBLOCK) = 0
brk(NULL) = 0xaab0bdc4000
brk(0xaab0bde9000) = 0xaab0bde9000
write(1, "Parent process: PID = 666, Child...., 43Parent process: PID = 666, Child PID = 667
") = 43
exit_group(0) = ?
+++ exited with 0 +++
harsha@Harsha:~/os-lab$
```

